

Tucker–ALS Update for Factor AA

We consider a Tucker parameterization of a 3D function

$$F(x_1, x_2, x_3) = \sum_{a,b,c} \sum_{i,j,k} G_{abc} A_{ai} B_{bj} C_{ck} T_{1i} T_{2j} T_{3k},$$

where A, B, C are factor matrices and G is the core tensor. In an ALS step we update A while B and C are fixed.

1. Merge the a –index

Define

$$\tilde{G}_{i(bc)} = \sum_a G_{abc} A_{ai}.$$

This is the mode–1 product $G \times_1 A$ reshaped so that (b, c) form a single index. The model becomes

$$F(x_1, x_2, x_3) = \sum_{i,j,k} \sum_{b,c} \tilde{G}_{i(bc)} B_{bj} C_{ck} T_{1i} T_{2j} T_{3k}.$$

2. Least–squares problem

Given sampled data of F , the merged matrix $\tilde{G}$ is obtained as the solution of a linear least–squares problem:

$$\tilde{G} = \underset{X}{\operatorname{argmin}} \|F - \operatorname{model}(X)\|_2^2,$$

where X has shape $(i) \times (bc)$.

3. SVD of the merged matrix

Reshape $\tilde{G}$ into a matrix and compute its SVD:

$$\tilde{G}_{i(bc)} = \sum_t U_{it} \sigma_t V_{(bc),t}.$$

4. Update of AA and the core

To enforce orthonormality of the factor matrix, set

$$A_{ai}^{\text{new}} = U_{ia}.$$

The core is updated by absorbing the singular values and right singular vectors:

$$G_{abc}^{\text{new}} = \sigma_a V_{(bc),a},$$

where $V_{(bc),a}$ is reshaped back into (b, c) indices.

5. ALS iteration

Cycle through the modes:

$$A \leftarrow \text{update}(A), \quad B \leftarrow \text{update}(B), \quad C \leftarrow \text{update}(C),$$

each time repeating the same procedure with the appropriate mode matricization. This is the classical Alternating Least Squares method for Tucker decomposition.